Project Name:- Retail Customer Retention Analytics - ADIDAS

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Customer Churn and Retention Analysis Project

Introduction

This project focuses on analyzing customer churn, retention, and lifetime value using Power BI. The analysis involves multiple datasets including customer demographics, transactions, loyalty programs, and store information. The objective is to derive actionable insights that help improve customer engagement, reduce churn, and increase overall customer lifetime value (CLV).

Problem Statement

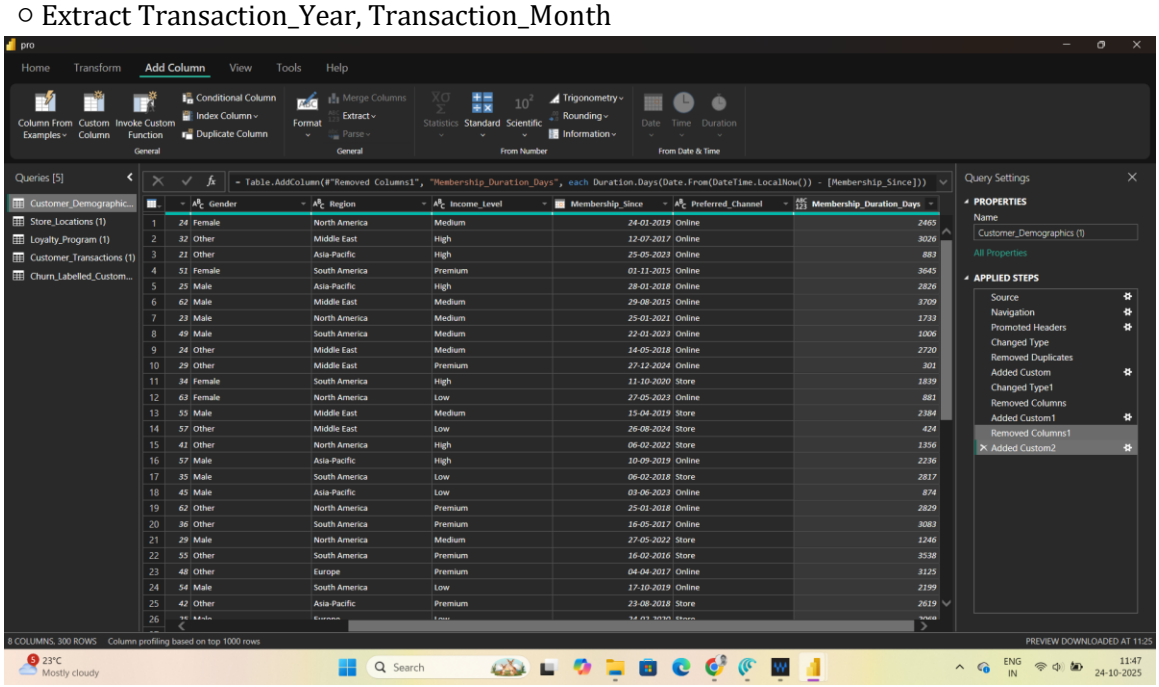
The primary goal is to identify factors influencing customer churn and retention across regions, income levels, loyalty tiers, and store types. Additionally, the project evaluates the impact of promotions, loyalty programs, and purchasing behavior on overall performance metrics.

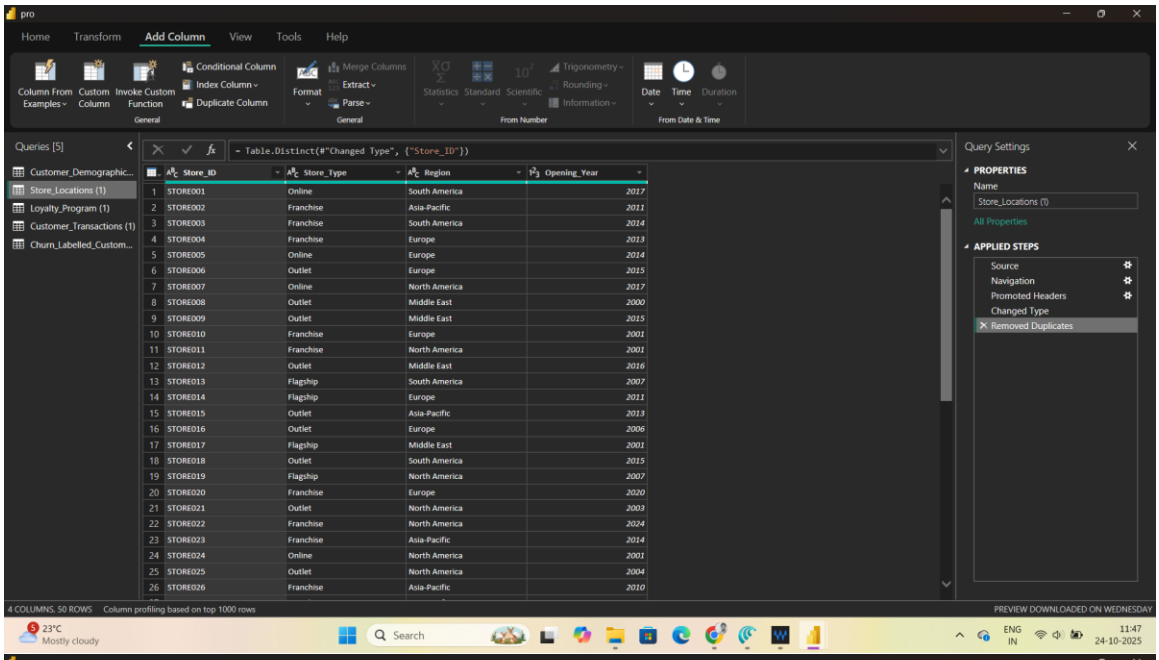
VIDEO LINK:-

https://drive.google.com/file/d/18Sg575yG0gArcLnVaenOdV9NuhqV60FP/view?usp=sharing

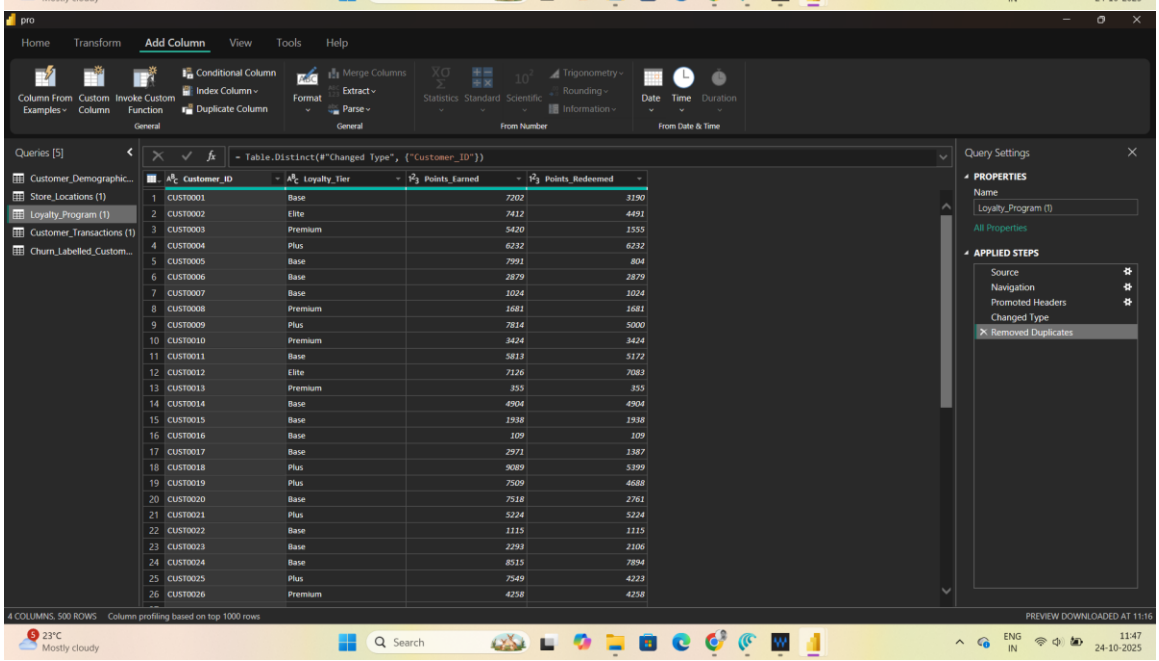
Task 1: Data Modeling & Cleaning (10 Marks)

- Load and transform datasets in Power Query
- Handle duplicates, missing values, and ensure correct data types
- Create calculated columns:
  - Membership_Duration = Today – Membership_Since
  - Extract Transaction_Year, Transaction_Month

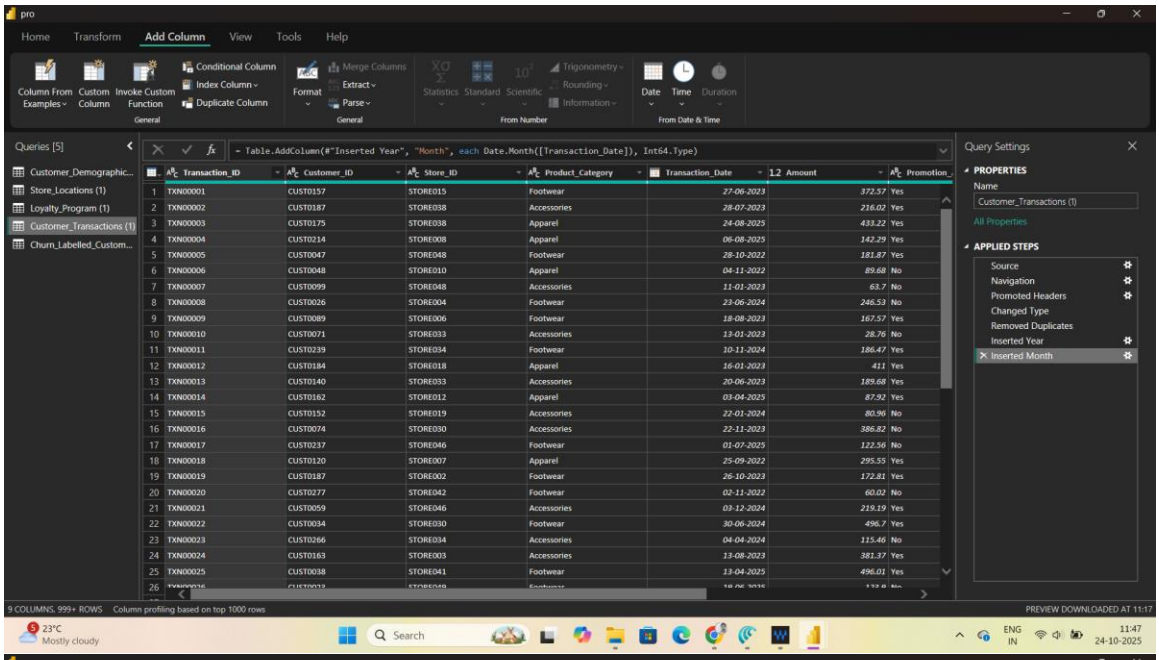


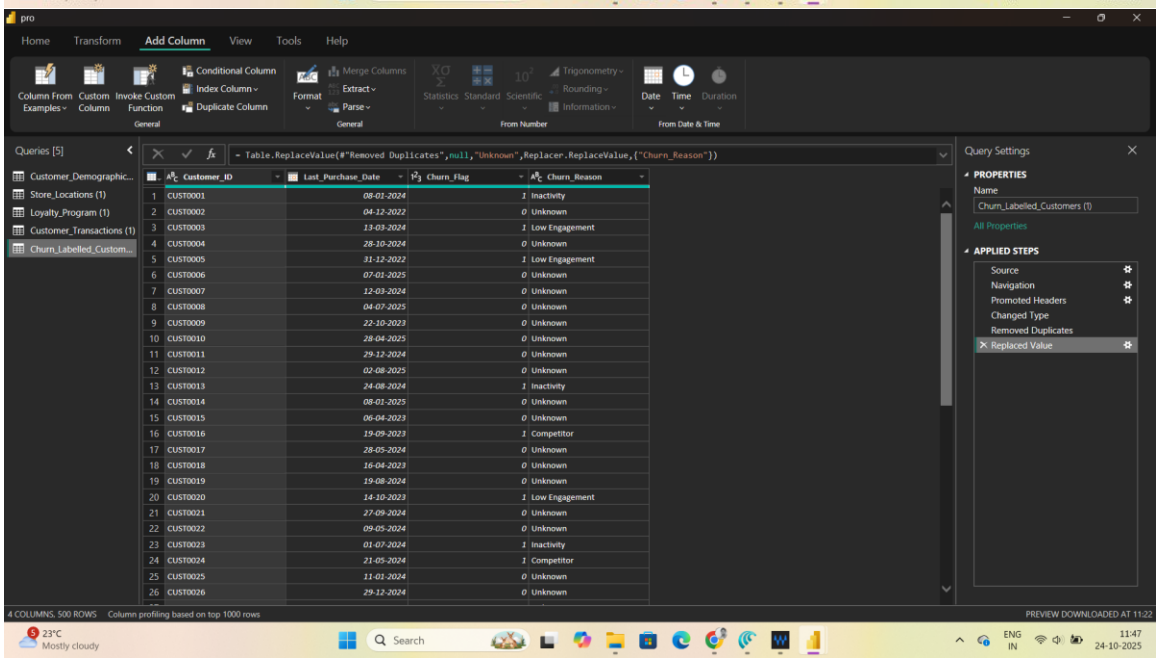


	Store_ID	Store_Type	Region	Opening_Year
1	STORE001	Online	South America	2017
2	STORE002	Franchise	Asia-Pacific	2011
3	STORE003	Franchise	South America	2014
4	STORE004	Franchise	Europe	2013
5	STORE005	Online	Europe	2014
6	STORE006	Outlet	Europe	2015
7	STORE007	Online	North America	2017
8	STORE008	Outlet	Middle East	2000
9	STORE009	Outlet	Middle East	2015
10	STORE010	Franchise	Europe	2001
11	STORE011	Franchise	North America	2001
12	STORE012	Outlet	Middle East	2016
13	STORE013	Flagship	South America	2007
14	STORE014	Flagship	Europe	2011
15	STORE015	Outlet	Asia-Pacific	2013
16	STORE016	Outlet	Europe	2006
17	STORE017	Flagship	Middle East	2001
18	STORE018	Outlet	South America	2015
19	STORE019	Flagship	North America	2007
20	STORE020	Franchise	Europe	2020
21	STORE021	Outlet	North America	2003
22	STORE022	Franchise	North America	2024
23	STORE023	Franchise	Asia-Pacific	2014
24	STORE024	Online	North America	2001
25	STORE025	Outlet	North America	2004
26	STORE026	Franchise	Asia-Pacific	2010



	Customer_ID	Loyalty_Tier	Points_Earned	Points_Redeemed
1	CUST0001	Base	7202	3190
2	CUST0002	Elite	7412	4491
3	CUST0003	Premium	5420	1555
4	CUST0004	Plus	6232	6232
5	CUST0005	Base	7991	804
6	CUST0006	Base	2879	2879
7	CUST0007	Base	1024	1024
8	CUST0008	Premium	1681	1681
9	CUST0009	Plus	7814	5000
10	CUST0010	Premium	3424	3424
11	CUST0011	Base	5813	5172
12	CUST0012	Elite	7126	7083
13	CUST0013	Premium	355	355
14	CUST0014	Base	4904	4904
15	CUST0015	Base	1938	1938
16	CUST0016	Base	109	109
17	CUST0017	Base	2971	1387
18	CUST0018	Plus	9089	5399
19	CUST0019	Plus	7509	4688
20	CUST0020	Base	7518	2761
21	CUST0021	Plus	5224	5224
22	CUST0022	Base	1115	1115
23	CUST0023	Base	2293	2106
24	CUST0024	Base	8515	7894
25	CUST0025	Plus	7549	4223
26	CUST0026	Premium	4258	4258



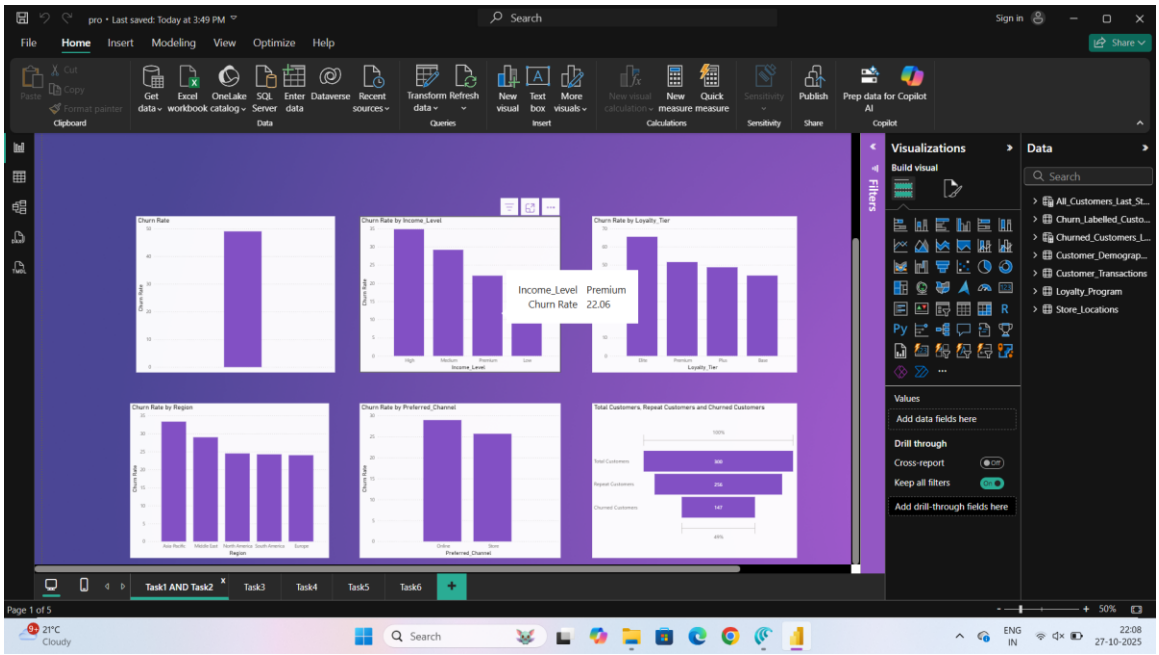


Key Findings & Insights:

While loading the Adidas datasets into Power BI, I found some duplicate records and a few missing values in the data. I used Power Query to remove these duplicates and replace or ignore missing entries, depending on what made sense for each field. I also made sure all columns had the correct data type (like date, number, or text). Then I created new calculated columns for things like membership duration and transaction year/month, which were super useful later for analysis.

Task 2: Churn & Retention Metrics (10 Marks)

- Create Churn Rate KPI = (Churned Customers / Total Customers) * 100
- Visualize churn rate by:
  - Region
  - Income Group
  - Channel (Store/Online)
  - Loyalty Tier
- Funnel Chart: Total Customers → Repeat Customers → Churned

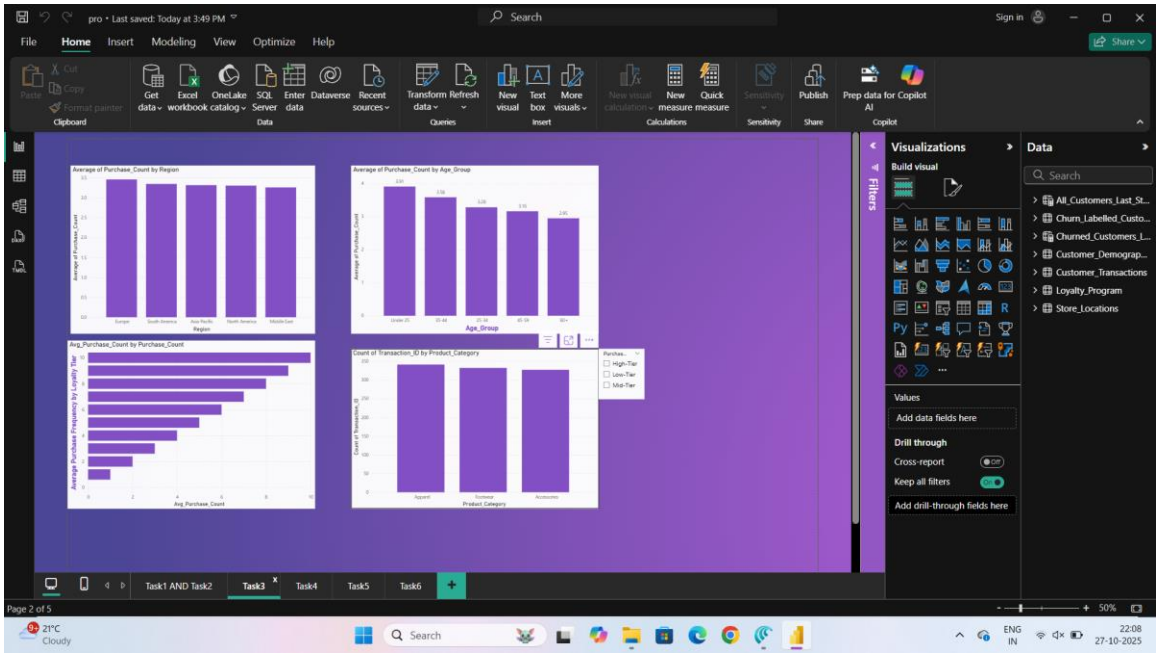


Key Findings & Insights:

My analysis showed that overall churn rate for Adidas customers is quite high. When breaking down the churn rate by regions, income levels, loyalty tiers, and channels, I noticed that some segments (like customers in certain regions or store types) were more likely to leave. The funnel chart also showed that only a portion of total customers are repeat buyers, and a significant share eventually churns. This means Adidas should keep a close watch on churn trends and focus on improving retention, especially in weaker segments.

Task 3: Repeat Purchase Analysis (10 Marks)

- Segment customers:
  - Low-Tier: 0–3 purchases
  - Mid-Tier: 4–8 purchases
  - High-Tier: 9+ purchases
- Compare avg. purchase frequency by Region, Age Group, Loyalty Tier
- Identify most purchased product categories by loyal customers

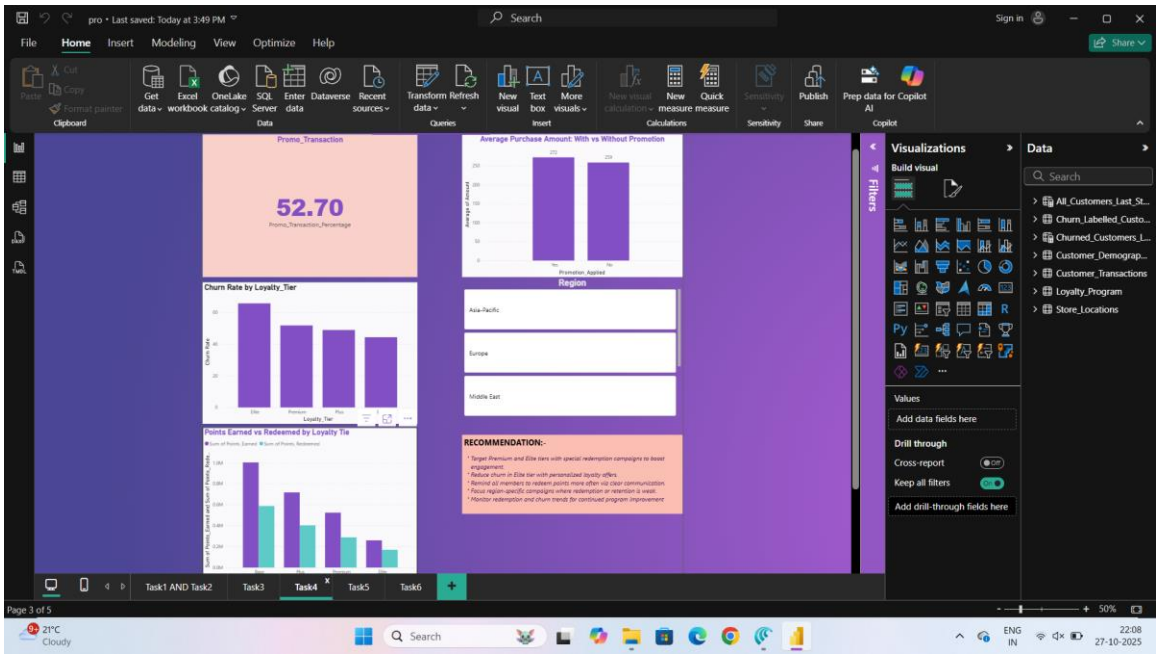


Key Findings & Insights:

Segmenting customers by number of purchases revealed that only a small percentage of customers reach the high-purchase (loyal) tier. Most customers are low or mid-tier. When I compared purchase frequency by region, age group, and loyalty tier, I found some interesting patterns (for example, certain regions and loyalty tiers have much higher average purchases). The most purchased product categories among loyal/high-tier customers are Apparel and Footwear.

Task 4: Promotion & Loyalty Impact (10 Marks)

- % of transactions with promotion applied
- Compare avg. purchase amount with vs without promotions
- Churn rate across loyalty tiers
- Points Earned vs Redeemed by Tier (clustered column chart)
- Recommendations to improve redemption & retention

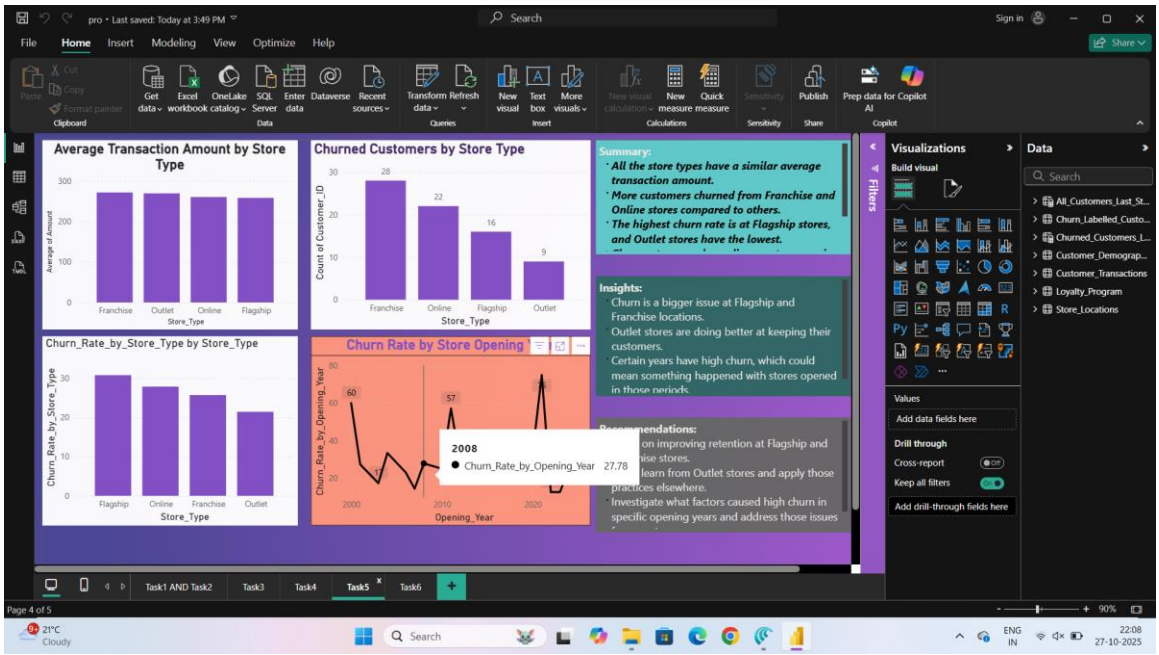


Key Findings & Insights:

I found that only around half of all transactions happen with a promotion applied. Customers who buy during promotions tend to spend a little more on average. Also, Premium and Elite loyalty tier members have higher retention and lower churn rates. The Points Earned vs. Redeemed chart showed some loyalty tiers are much better at redeeming points, which probably makes them more engaged with the brand. Overall, strong loyalty program engagement is linked to higher retention.

Task 5: Store & Channel Performance vs Retention (10 Marks)

- Merge store data with transactions
- Visualize:
  - Avg. transaction amount by Store Type
  - Churn rate by store type
  - Correlation between store opening year & retention

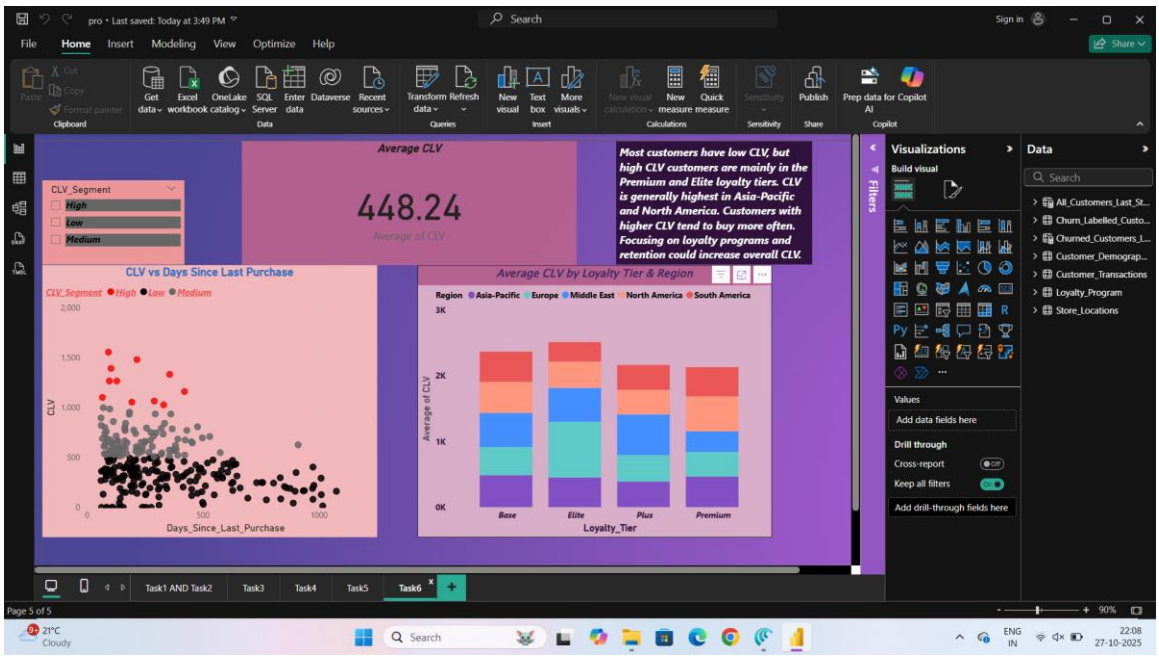


Key Findings & Insights:

After combining store data with transactions, I saw that average transaction amounts are quite similar across store types but churn rates vary a lot. Franchise and Flagship stores have higher churn, while Outlet stores perform best at retaining customers. The churn rate also fluctuates based on store opening year—some years stand out with much higher customer loss, which might be due to unique events or changes in those periods.

Task 6: Customer Lifetime Value (CLV) Analysis (10 Marks)

- CLV = Total Amount Spent / Membership Duration (Years)
- Segment customers into Low, Medium, High CLV
- Visualize:
  - CLV vs Days Since Last Purchase
  - CLV by Loyalty Tier & Region



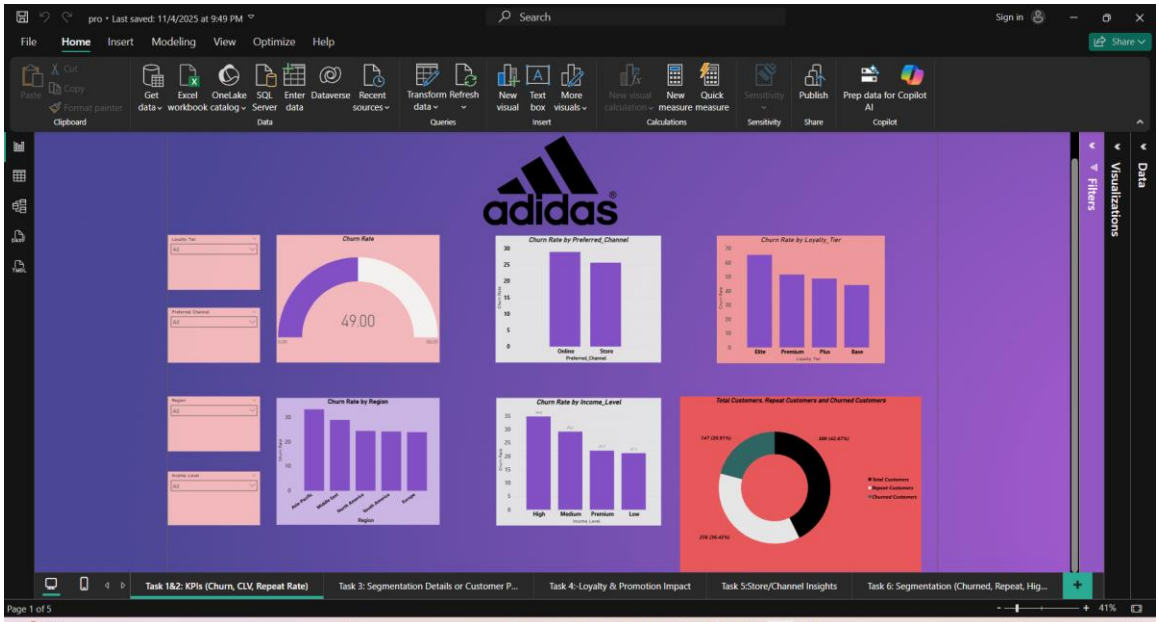
Key Findings and Insights

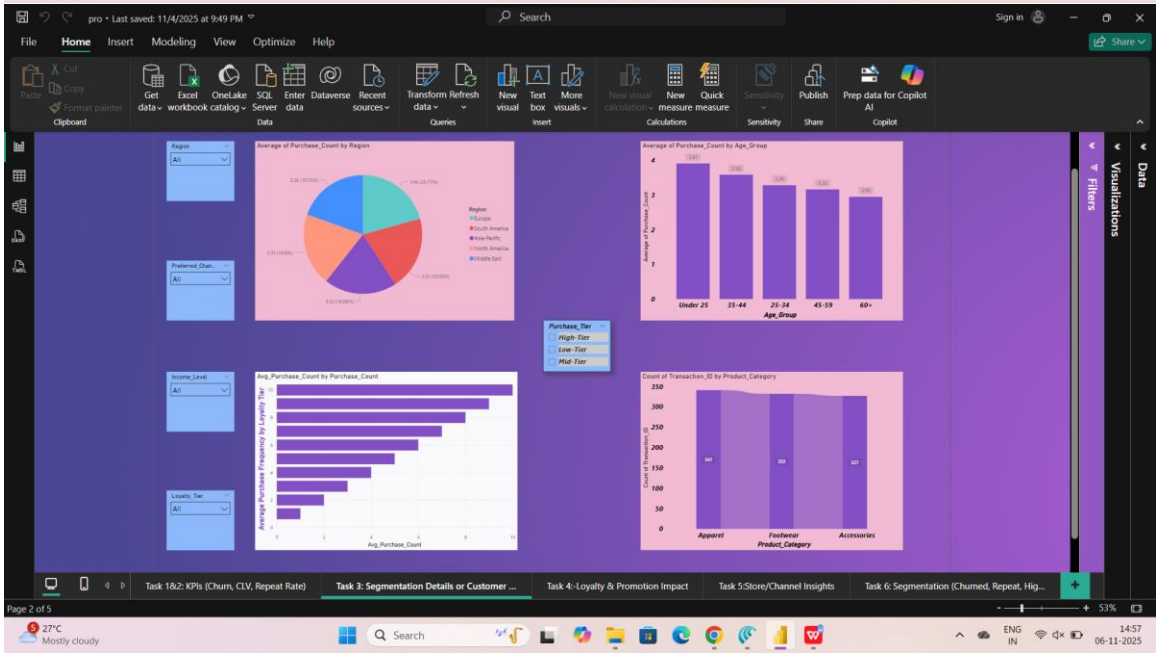
- Higher churn rates are observed among Premium and Elite loyalty tiers.
- Promotion campaigns positively influence repeat purchase frequency.
- Store performance varies, with Flagship stores showing higher churn.
- Customers with higher CLV are generally part of higher loyalty tiers and regions with strong retention programs.

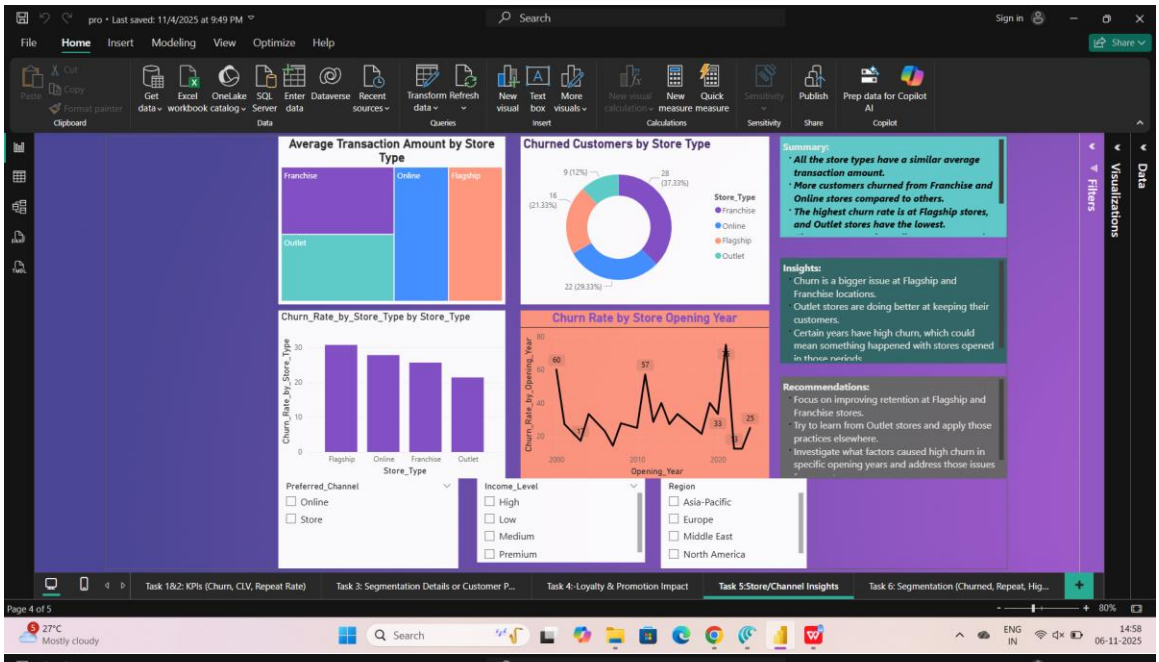
Task 7: Final Dashboard & Executive Summary (20 Marks)

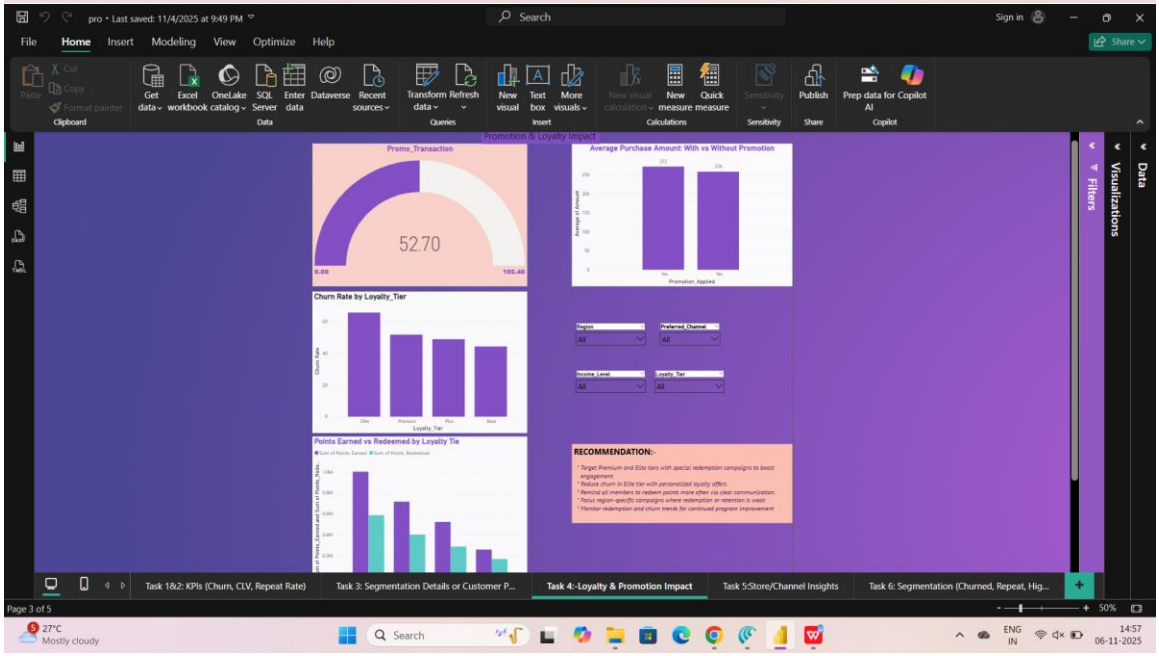
● Multi-page Power BI Report:

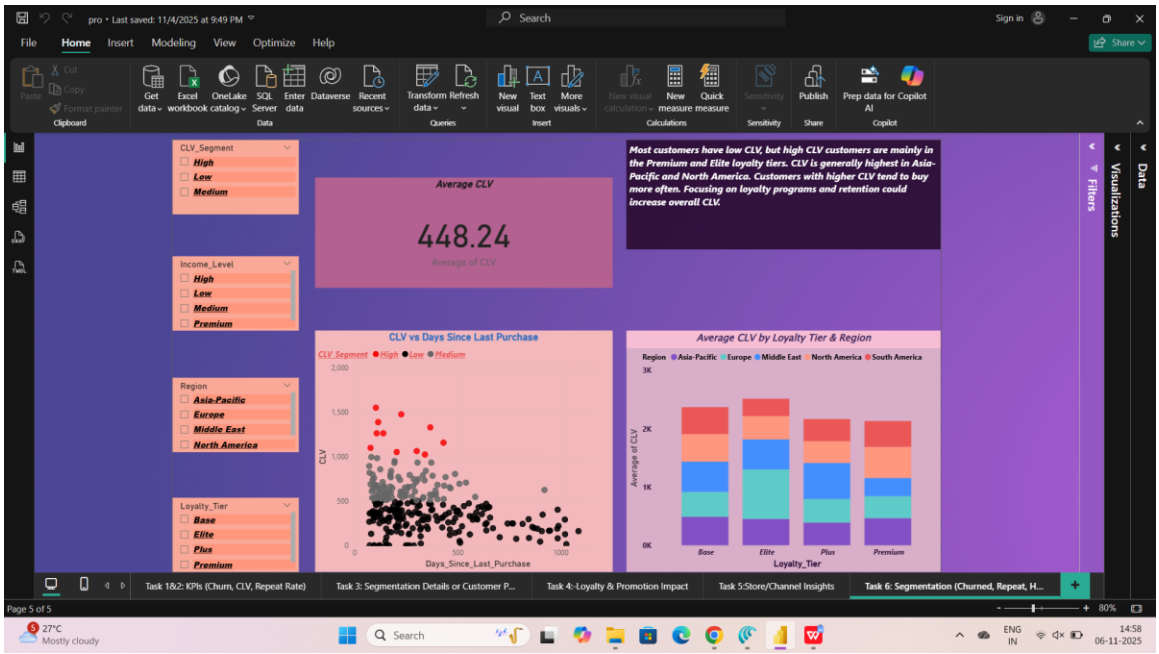
○ Page 1: KPIs (Churn, CLV, Repeat Rate) ○ Page 2: Loyalty & Promotion Impact ○ Page 3: Store/Channel Insights ○ Page 4: Segmentation (Churned, Repeat, High-Value) ● Slicers: Region, Channel, Income, Loyalty Tier






Average Transaction Amount by Store Type
Franchise
Online
Flagship
Outlet
Churned Customers by Store Type
9 (12%)
28 (37.33%)
16 (21.33%)
22 (29.33%)
Store_Type
Franchise
Online
Flagship
Outlet
Summary:
All the store types have a similar average transaction amount.
More customers churned from Franchise and Online stores compared to others.
The highest churn rate is at Flagship stores, and Outlet stores have the lowest.
Insights:
Churn is a bigger issue at Flagship and Franchise locations.
Outlet stores are doing better at keeping their customers.
Certain years have high churn, which could mean something happened with stores opened in those periods
Recommendations:
Focus on improving retention at Flagship and Franchise stores.
Try to learn from Outlet stores and apply those practices elsewhere.
Investigate what factors caused high churn in specific opening years and address those issues
Churn_Rate_by_Store_Type by Store_Type
Churn_Rate_by_Store_Type
Flagship
Online
Franchise
Outlet
Store_Type
Churn Rate by Store Opening Year
Churn_Rate_by_Opening_Year
Opening_Year
Preferred_Channel
Online
Store
Income_Level
High
Low
Medium
Premium
Region
Asia-Pacific
Europe
Middle East
North America
Task 1&2: KPIs (Churn, CLV, Repeat Rate)
Task 3: Segmentation Details or Customer P...
Task 4:-Loyalty & Promotion Impact
Task 5:Store/Channel Insights
Task 6: Segmentation (Churned, Repeat, Hig...
Page 4 of 5


Promotion & Loyalty Impact
Promo_Transaction
52.70
Average Purchase Amount: With vs Without Promotion
Churn Rate by Loyalty_Tier
Points Earned vs Redeemed by Loyalty Tie
RECOMMENDATION:-
Target Premium and Elite tiers with special redemption campaigns to boost engagement
Reduce churn in Elite tier with personalized loyalty offers
Remind all members to redeem points more often via clear communication
Focus region-specific campaigns where redemption or retention is weak
Monitor redemption and churn trends for continued program improvement
Task 1&2: KPIs (Churn, CLV, Repeat Rate)
Task 3: Segmentation Details or Customer P...
Task 4:-Loyalty & Promotion Impact
Task 5:Store/Channel Insights
Task 6: Segmentation (Churned, Repeat, Hig...
Page 3 of 5



1. Which customers should Adidas prioritize for retention?

From my analysis, Adidas should focus most on retaining customers in the Premium and Elite loyalty tiers, especially those who have a high Customer Lifetime Value (CLV). These customers tend to buy more often and spend more in total. If Adidas keeps these top-tier customers happy and loyal, it will have the biggest impact on sales and profit.

2. Which channels are underperforming?

Looking at the data, I noticed that churn rates are highest at Flagship and Franchise stores. These channels are losing more customers than others and need attention. Also, while online sales are good, there's room to boost repeat purchases there. I recommend Adidas work on improving service, communication, and special retention offers in Flagship, Franchise, and online channels.

3. How can Adidas strengthen loyalty program engagement?

I found that loyalty program members who get more personalized offers and can easily redeem their points are more likely to stick with Adidas. To make the loyalty program even better, Adidas should:

Give special offers to Premium and Elite members,

Make point redemption simpler for everyone,

Use personalized messages and rewards to keep all loyalty members engaged and encourage them to move to higher tiers.